Object Detection

Introduction

Object Detection is a computer vision task that identifies objects within an image and determines their locations using bounding boxes.

Unlike image classification, object detection answers:

- What object is present?
- Where is it located?

Object Detection Pipeline

Input Image

↓

Feature Extraction

↓

Region Proposal / Grid Prediction

↓

Classification

↓

Bounding Box Regression

↓

Final Detection

Bounding Box

A bounding box consists of:

- x-coordinate
- y-coordinate
- width
- height

Confidence score:

Example:

Person – 98%

Dog – 94%

Car – 87%

Popular Models

YOLO (You Only Look Once)

Advantages:

- Extremely fast
- Real-time detection
- High accuracy

Applications:

- Autonomous driving
- CCTV
- Robotics

Faster R-CNN

Advantages:

- Very accurate
- Two-stage detector

Disadvantage:

- Slower than YOLO

SSD (Single Shot Detector)

Balances speed and accuracy.

RetinaNet

Introduced Focal Loss to improve detection of rare objects.

Evaluation Metrics

IoU (Intersection over Union)

Measures overlap between predicted and actual boxes.

IoU = Intersection / Union

---

mAP (Mean Average Precision)

Standard benchmark metric for object detection models.

Applications

- Self-driving cars
- Face Detection
- Medical Imaging
- Retail Analytics
- Traffic Monitoring
- Security Surveillance
- Drone Navigation

Conclusion

Object Detection combines classification and localization, enabling AI systems to understand complex visual scenes.